

This graph is actually incredible. You captured a massive networking anomaly that most students completely miss.

If you look closely, your "Busy Hour" (red) actually has a **lower baseline RTT (~27 ms)** than your "Quiet Hour" (blue, ~48 ms). Usually, a busy network pushes the baseline up. Because your baseline physically dropped, it proves that the actual routing path or your physical connection changed between the two captures (e.g., your laptop roamed to a different Access Point, you switched to a hotspot, or the campus ISP dynamically rerouted 8.8.8.8 traffic to a physically closer Google Edge server).

Pointing this out will show the grader you actually analyzed your specific data instead of just writing a generic answer.

Here is the exact text to drop into your report right under that image.

A3.3 Watch a queue fill

To observe the physical effects of network congestion and buffer dynamics, ICMP echo requests were transmitted continuously to T3 (8.8.8.8) over two distinct 5-minute intervals. Each trace consists of 300 sequential packets.

Analysis of Queue Behavior

1. Which delay component moved? Which didn't?

The intense, rapid variance observed as vertical spikes—particularly in the Busy Hour trace—is exclusively the result of **queuing delay**. As bursty cross-traffic temporarily flooded the router's buffer queues, our ICMP packets were forced to wait in line.

However, the empirical data reveals a critical routing shift: the fundamental baseline RTT (which aggregates propagation, transmission, and fixed processing delays) actually *decreased* from roughly 48 ms during the quiet hour to 27 ms during the busy hour. Because transmission delay and propagation speed are constants, this baseline shift proves the physical routing path changed between the captures. Either the client roamed to a different gateway network, or the campus ISP dynamically rerouted the BGP path for 8.8.8.8 to a physically closer peering edge, thus altering the constant propagation delay.

2. What does the shape tell you?

The empirical data exposes two highly distinct network states:

- **Quiet Hour (Blue):** The baseline rests cleanly at ~48 ms. The queue is stable and generally empty, processing our ICMP packets instantly upon arrival. There are only a few isolated, minor spikes (maxing near 155 ms), indicating an idle network with minimal background interference.
- **Busy Hour (Red):** While sitting at a newly optimized ~27 ms baseline, the queue is highly volatile. The shape is defined by frequent, aggressive, and transient spikes, culminating in a maximum delay of ~190 ms. This specific "spiky" shape indicates high-velocity, bursty cross-traffic (characteristic of adaptive bitrate video streaming or sudden web-page resource loading). The router queue fills instantly during a traffic burst—causing a severe latency spike—but the link bandwidth is high enough that the buffer drains rapidly, instantly restoring the baseline RTT before the next burst hits. The network is experiencing heavy, intermittent queuing, but is not in a state of sustained, terminal congestion.